

SMALL HOSPITAL EMR SYSTEM

Modern Cloud-Based Electronic Medical Records

For 50-200 Bed Private Hospitals



Complete Implementation & Deployment Guide

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EXECUTIVE SUMMARY

Our Small Hospital EMR is a modern, cloud-based electronic medical records system designed specifically for private hospitals with 50-200 beds. It integrates all hospital departments on a single platform, reducing costs by 90%, implementing in just 12 weeks, and eliminating the need for a dedicated IT team.

Key Highlights

- ✓ 12 Fully Integrated Modules
- ✓ 70+ Beautiful, Responsive Pages
- ✓ 150+ REST API Endpoints
- ✓ Cloud-Based Stack (Supabase + Koyeb + Vercel)
- ✓ No IT Team Required
- ✓ Only \$200-300/month Operating Costs
- ✓ 12 Weeks from Order to Live System
- ✓ HIPAA-Compliant Design
- ✓ You Own It
- ✓ Scalable (50 to 500+ beds)

WHAT IS THIS EMR SYSTEM?

The Small Hospital EMR is a complete, integrated healthcare management platform that brings all hospital departments onto a single digital system. It replaces paper records, manual processes, and disconnected spreadsheets with an intelligent, secure, and easy-to-use web application.

Core Characteristics

Modern Solution: Cloud-based system built with latest technology

Complete Integration: All departments connected

Easy to Use: Intuitive interface for busy clinicians

Scalable Infrastructure: Grows with your hospital

Affordable: Cloud stack keeps operating costs low

Secure & Compliant: Bank-grade encryption, HIPAA design

Fast Deployment: Go live in 12 weeks vs 2 years

Owned by You: You own the code and data

COMPLETE FEATURE OVERVIEW: 12 MODULES

1. PATIENT MANAGEMENT

- New patient registration with demographics
- Unique patient ID (UHID) automatic generation
- Complete medical history tracking
- Allergy & clinical alert management
- Document upload (ID cards, insurance, referrals)

2. ELECTRONIC MEDICAL RECORDS (EMR)

- SOAP consultation notes
- Diagnosis recording with ICD-10 coding
- E-prescribing with real-time drug interaction checking
- Vital signs recording & auto-calculations
- Specialty-specific EMR templates

3. APPOINTMENTS & SCHEDULING

- Online appointment booking (24/7 self-service)
- Walk-in queue management with real-time waiting times
- Doctor availability management & scheduling
- Automated appointment reminders (SMS + Email)
- Telemedicine slot booking

4. OUT-PATIENT DEPARTMENT (OPD)

- Patient check-in interface
- Queue management & display system
- Doctor consultation workflow
- Specialist referral management
- OPD performance reports & analytics

5. PHARMACY MANAGEMENT

- Automatic e-prescription receipt from EMR
- Medication dispensing interface
- Real-time drug interaction checking
- Inventory management (stock levels, alerts)

- Barcode-based medication dispensing

6. LABORATORY (LIS - Lab Information System)

- Digital laboratory order receipt
- Sample collection tracking with barcodes
- Result entry interface & auto-validation
- Critical value alerts (automatic notification)
- Turnaround time (TAT) monitoring

7. IN-PATIENT / WARDS MANAGEMENT

- Admission workflow & documentation
- Bed management with visual bed occupancy map
- Patient transfer tracking between wards
- Discharge planning & documentation
- Daily census reports

8. NURSING STATION

- Patient worklist (assigned patients display)
- Vital signs charting & trend monitoring
- Care plan creation & management
- Medication administration tracking (eMAR)
- Real-time synchronization with doctors

9. BILLING & FINANCIAL MANAGEMENT

- Automatic billing from consultation notes
- Invoice generation & printing
- Multiple payment method support
- Insurance claim submission
- Revenue tracking by department

10. ADMIN DASHBOARD

- Real-time KPI dashboards
- Revenue monitoring & trending
- Patient statistics & demographics
- Department performance metrics
- Automated record export (CSV, PDF)

11. USER MANAGEMENT & RBAC

- 7 predefined roles (Doctor, Nurse, Pharmacist, Lab Tech, etc)
- Role-based access control (RBAC)
- Granular permission management
- Multi-factor authentication (MFA) support
- Session timeout & security policies

12. SECURITY & AUDIT TRAIL

- Complete audit logging (every action tracked)
- Data encryption (AES-256 at rest, TLS 1.3 in transit)
- Automated daily backups with 30-day retention
- Disaster recovery with point-in-time recovery
- HIPAA-compliant architecture & controls

SYSTEM ARCHITECTURE

The Small Hospital EMR is built on a modern, scalable cloud architecture using three leading managed services. This eliminates the need for a dedicated IT team while providing enterprise-grade reliability and security.

Three-Tier Architecture

Frontend (Vercel)	Next.js 14 + React + TypeScript 70+ pages, real-time, PDF/CSV export
Backend API (Koyeb)	Node.js + NestJS + TypeScript 150+ endpoints, auto-scaling (1-5 instances)
Database (Supabase)	PostgreSQL + Supabase Auth 80+ tables, Row-Level Security, auto-backups

Why This Stack?

- ✓ No DevOps Team Required
- ✓ 99.9% Uptime SLA
- ✓ Auto-Scaling
- ✓ Cost-Effective (\$200-300/month)
- ✓ Fast Deployments
- ✓ Enterprise Security
- ✓ Scales from 50 to 500+ beds

KEY BENEFITS & RETURN ON INVESTMENT

60% Wait Time Reduction	Queue management + streamlined workflows
40% Billing Efficiency	Automated invoicing & claim submission
70% Fewer Medical Errors	Drug interaction checks, alerts
90% Paper Elimination	Fully digital records
25% Revenue Increase	Accurate billing, fewer rejections
50% Staff Productivity Gain	Automation of repetitive tasks

Cost Savings Analysis

Metric	Our System	Enterprise	Savings
Setup Cost	\$5-13K	\$100-500K	\$95-487K
Monthly Cost	\$200-300	\$5-10K+	\$4,700-9,800
Annual Cost	\$2,400-3,600	\$60-120K+	\$56,400-116,400
Year 1 Total	\$7-16K	\$160-600K+	\$153-593K

IMPLEMENTATION TIMELINE: 12 WEEKS

Week	Phase	Deliverable
1	Architecture & Setup	Design, database schema, API spec
2-3	Backend Development	150+ endpoints, 80+ tables
4-5	Frontend Development	70+ pages, responsive design
6	Testing & Integration	E2E testing, security testing
7	Deployment	Supabase, Koyeb, Vercel setup
8	Documentation	API docs, user guides, training
9-12	Refinement & Go-Live	User feedback, training, launch

■ Result: PRODUCTION-READY EMR SYSTEM LIVE

PRICING & COST STRUCTURE

One-Time Development Cost

Service	Cost
Complete System Development	\$5,000 - \$10,000
Data Migration (if applicable)	\$1,000 - \$3,000
Staff Training & Support	Included
TOTAL DEVELOPMENT	\$5,000 - \$13,000

Monthly Operating Costs

Service	Cost
Supabase Database	\$25 - \$250
Koyeb Backend	\$20 - \$200
Vercel Frontend (CDN)	\$20 - \$100
Email & SMS Services	\$10 - \$50
Domain & SSL	\$10 - \$20
TOTAL MONTHLY (Small Hospital)	\$200 - \$300

SECURITY & COMPLIANCE

Data Protection

- AES-256 Encryption (data at rest)
- TLS 1.3 Encryption (data in transit)
- ✓ Multi-factor Authentication (MFA)
- ✓ Role-Based Access Control (RBAC)
- ✓ Row-Level Security (RLS) database policies
- Complete audit logging (all actions tracked)
- Daily automated backups (30-day retention)

Regulatory Compliance

- HIPAA Design Compliance
- GDPR Support (data export, right to be forgotten)
- Healthcare Data Security Standards
- Session timeout & auto-logout (30 minutes)
- Encrypted password storage (bcrypt)
- SSL/TLS certificates (auto-renewal)
- 24/7 security monitoring & alerts

SUPPORT & TRAINING PROGRAM

Training Materials Provided

- ✓ 20+ video tutorials (step-by-step)
- ✓ User guide (50+ pages, PDF)
- ✓ Administrator guide (configuration)
- ✓ Troubleshooting guide (common issues)
- ✓ Quick reference cards (laminated)
- ✓ Role-specific training slides

Support Services Included

- ✓ 24/7 Support (go-live week)
- ✓ Email support (ongoing)
- ✓ Bug fixes (within 24 hours)
- ✓ Automatic security updates
- ✓ Monthly check-ins (first 3 months)
- ✓ Performance optimization

Service Level Agreement (SLA)

- 99.9% Uptime Guarantee (max 43 min downtime/month)
- < 1 Hour Response Time (critical issues)
- < 24 Hour Resolution (bugs fixed and deployed)
- Daily Automatic Backups (zero data loss)
- Auto-Updates & Security (patches applied automatically)

COMPETITIVE COMPARISON

Feature	Our Solution	Enterprise EMR	Advantage
Setup Cost	\$5-13K	\$100-500K	Save \$95-487K
Monthly Cost	\$200-300	\$5-10K+	Save \$4,700-9,800
Implementation	12 weeks	2+ years	20 months faster
All-in-One Modules	YES	Multiple vendors	Cost savings
Ease of Use	Doctor-friendly	IT-focused	Less training
No IT Team Needed	YES	NO	Save \$150-300K/year
Data Ownership	You own it	Proprietary	Full control

NEXT STEPS: LET'S GET STARTED

Ready to transform your hospital with a modern, affordable EMR system? Here's how to get started:

1. Schedule a Discovery Meeting
2. Confirm Project Scope
3. Sign Agreement
4. Development Begins (Week 1)
5. Weekly Check-ins
6. 12 Weeks Later: Go Live
7. Ongoing Support

Contact us to schedule your discovery meeting

Ready to go live in 12 weeks!